

Sachin Kumar

PCB Hardware Design Engineer

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Professional Summary:

Passionate Hardware/PCB Design Engineer with 8.8 years of Experience in Schematic design , Circuit Simulations and Multi-layer complex PCB Layouts. Skilled in component engineering, Library creation and Technical documentation, with Specialization in high speed/mixed signal circuit design, compliance driven product development, and cost optimized solutions.

Professional Experience:

Capgemini Technology Services India Ltd

Pune, Maharashtra, India

Professional II (ER&D)

June 2024 – Present

- Performed **technical analysis of obsolete components and 8-layer PCB design changes** in medical products and prepared supporting documentations.
- Selected **analog/digital alternate components** considering **lifecycle, RoHS & REACH compliance**, and material composition to protect from risk posed by chemical.
- Developed test-plans, test-cases, and reports for design changes as per regulatory standards.
- **Led parametric, functional, EMC/EMI, and safety testing as per IEC medical standards** in internal and external labs.
- **Reduced cost and lead time of components**; reviewed **schematics and PCB/Gerber layouts** against legacy designs.
- Collaborated with **Japanese and US global teams** for technical discussions, **component finalization**, and **technical assessment of obsolete vs. alternate parts**.
- Managed **pre-compliance testing** as per **IEC customized standards** from TÜV Rheinland Lab.

Anand Group India (ANHS & AAPL)

Gurugram, Haryana, India

Assistant Manager (R&D)

July 2022 – June 2024

- **Benchmarked** competitor products, **conducted market research**, and analyzed **customer SOR/technical specifications**.
- Performed **design calculations, power budget analysis, E-BOM and Assembly BOM preparation, cost reduction**, and **vendor/supplier coordination** for new components/technologies.
- Executed **schematic capture, library creation, multi-layer PCB design (as per IPC standards), PDN analysis, Gerber review**, and **PCB manufacturing/assembly coordination**.
- **Led board bring-up, testing, debugging, RCA/WCA, prototype development, and full design cycle**.
- Collaborated with supply chain, mechanical, and firmware/software teams.
- Designed **high-frequency magnetics and SiC semiconductor based 750W AC-DC converter achieving 95% efficiency** in a small form factor with **IP66, 0.99 power factor and ATHD less than 10%**; also developed **low-cost isolated AC voltage sensing circuits**.
- Designed **500W AC-DC Power converter with PFC/LLC Resonant converter without auxiliary supply**.
- Conducted hardware troubleshooting, **DFMEA, DFM/DFT**, and prepared **DVP test plans**.
- Managed and led the hardware/testing teams, **set up R&D labs with required test equipment's**, and oversaw product certification with IEC approvals from accredited labs.

EFILL Electric Vehicle & Charging Solutions Private Limited

Sonipat, Haryana, India

Senior Hardware Engineer (R&D)

Dec 2020 - June 2022

- Conducted competitor's product **benchmarking, market research**, and **customer specification analysis**.
- Performed **design calculations, power budget analysis, E-BOM and Assembly BOM preparation, cost reduction**, and **vendor/supplier coordination** for new components/technologies.
- **Reduced the cost** by selecting the electronic, electrical components, switchgears like **MCB, MCCB, RCCB, Relay contactors, CT Coils, 1- core multistrand wires, 3 & 5-core cables** for **Fast DC EV charging Stations rating 15kW/30kW/60kW** with same specifications.

- Executed **schematic capture, library creation, multi-layer PCB design, PDN/signal integrity analysis**, Gerber review, and PCB manufacturing/assembly coordination with PCB Manufacturer.
- Integrated **GSM (4G/3G/2G), Wi-Fi/BLE, Ethernet, NFC, RS232/RS485, CAN**, and **keypad matrix interfaces** with **32-bit ARM Cortex M4 microcontroller**.
- Designed **high-precision analog sensing**, relay control, **LDOs, DC-DC converters, and isolated circuits**.
- Led circuit simulation, debugging, RCA/WCA, and prepared test plans, test-cases, and test reports.
- Collaborated with **mechanical, supply chain, firmware/software, sales, I&C teams, and local & global suppliers**.
- Integration with **Centralized Software CMS (OCPP with 4G, Wi-Fi and Ethernet** and Real Time Mobile Android app Testing based on OCPP1.6J).
- **Built test jigs/benches** for **component/system-level validation with vehicles in R&D lab**.
- **Reduced product cost and form factor** by **alternate component selection** and technical methodology.
- Designed **wire/harness diagrams** and coordinated with vendor for development/manufacturing.
- Support production and quality department for manufacturability, testability of the products on production line.
- Designed and tested **2kW and 3.3kW Off-Board AC-DC Charger for L3 E-Rickshaw and L5 E-Auto** with PFC and less than 10% ATHD.
- Tested and Validated **7.4kW/15kW/30kW AC, DC EV Fast Charging stations** with CMS and mobile applications to fulfil the Vehicle SOC level.

Kreate Lights Private Limited

Noida, Uttar-Pradesh India

R&D Engineer

Feb 2020 -Dec 2020

- Designed and developed **Astronomical Timer, UV Timer, and microwave sensor-based controllers**.
- Created **schematics in Altium designer, selected components, and prepared test benches** for system/component-level testing.
- Managed **sourcing/procurement** of **critical components** and supplier technical discussions.
- Oversaw product certification with IEC compliance from NABL-accredited labs.
- Performed **heat sink calculations, design drawings using SolidWorks**, and sample development with supplier for power devices.
- Worked on **Smart Pole CCMS based on IOT with Weather Sensor (PM2.5/PM10 Sensor/Temperature, Humidity, etc), Broadcasting Audio/Video, Smart LED Lights for DRDO**.
- Tested, troubleshoot, and calibrated **3.7kW single-phase AC Type 2 EV charging station (Phoenix Contact)**.

Protonix Fortuner India Private Limited

Noida, Uttar-Pradesh India

R&D Engineer

June 2019 – Feb 2020

- Designed Schematic and **2-Layer PCB layout** and developed **PWM/MPPT-based solar charge controllers for LED lights with led dimmable (0-100%) feature and PIR motion sensor for Philips, Halonix, Syska, etc**.
- Developed **voltage/current sensing circuits, LED drivers, LCD/PIR sensor interfaces** with microcontrollers.
- Created **schematics and 2-layer PCB layouts in Altium**; selected components and managed PCB design.
- Designed and tested **18W AC/DC emergency tube lights and AC-DC LED drivers/lights (24W, 30W, 60W)** for many OEMs.
- Prepared **CKD/SKD BOMs**, test procedures, reports, ECN/ECR documentation.
- Debugged and **reverse-engineered LED driver PCBs** for compatibility and quality assurance.
- Designed and developed **magnetics (transformers/inductors) based on specifications and ratings with vendor**.

Midas Private Limited

Noida, Uttar-Pradesh India

R&D Engineer

Dec 2018 – June 2019

- Designed the **schematic diagram and 2-layer PCB layout on Eagle CAD for Bluetooth/Infrared sensor-based LED control devices**.
- Tested and debugged LED drivers with different power ratings and package; also prepared technical documentation.
- Reverse-engineered and developed **Bluetooth/Infrared sensor-based LED control devices**.
- Worked on **government tenders** to meet technical requirements like LM79, LM80, photo- biological safety test reports.
- Calculated lux levels and selected appropriate indoor/outdoor lighting solutions.
- Designed indoor/outdoor lighting using DIALux evo software.

- Created **symbols/footprints, schematics, and 1-Layer PCB layouts on Eagle CAD for different kind of Analog/digital Sensors, Motor Driver, 7-segment matrix, Alphanumeric LCD Display, breakout board and so on.**
- **Interfaced sensors, LCD, keypad, Bluetooth/Wi-Fi modules, and dc motor drivers with 8-bit microcontrollers;** tested and debugged hardware modules, evaluation board/training kits.
- Given training to college students based on Arduino/AVR/Raspberry Pi and other robotics kits.
- Responsible for robotics lab establishment and exhibition in colleges.
- Got two times **appreciation** from the customer on providing excellent services
- Prepared technical documentation, user manuals for learning kits.

Education Background:

Qualifications	School/College	University	Duration	Percentage
B. Tech (ECE)	Delhi Institute of Engineering & Technology, Meerut	AKTU	2013-2017	76% (hon's)
12th (PCM)	Jawahar Navodaya Vidyalaya, Meerut	CBSE	2012-2013	67.2 %
10th (Science & Maths)	Jawahar Navodaya Vidyalaya, Meerut	CBSE	2010-2011	9.0 CGPA

Technical Skills:

Software/Tools: Altium Designer, Eagle CAD, OrCAD Capture, Allegro PCB Designer, Multisim, Proteus, SolidWorks, Saturn PCB Toolkit, Hyperlynx,

Design Competency: High-Speed Design, Mixed Signal Design, Power Electronics, RF Design, PCB Designing, Component Engineering, Cost reduction, Thermal Management, Magnetics Design, EMI/EMC Design, SI/PI analysis.

Power Topology: Buck/Boost converter, Flyback Converter, LLC Resonant Half Bridge/Full Bridge, Synchronous Rectifier, LDO, PMIC Power Rails, EMI filter.

Standards and regulations: IEC-61851-1/2, IS-17017-1/2, AIS-138-1/2, IEC-60601-1-2, IEC-61000-3-3, IEC-61000-4-3, IEC-61000-4-5, IEC-61000-4-6, CISPR-11/22/25.

Participations & Certifications:

- Life-Science Medtech L1 and L2 Certifications from Capgemini
- Architect L0 Certification from Capgemini
- Certification-Emerging Leadership Program from Capgemini in collaboration with Harvard Business Education
- Advanced High-Speed Hardware and PCB Design Training/certification from Udemy
- High-Speed Board Design – SOM Training/certification from Udemy
- Mixed Signal Hardware & PCB Design Training /certification from Udemy
- Certification-Industrial Training in 'Embedded System by Arduino' from CETPA Noida
- MATLAB/Simulink Certification from Udemy
- Participation in Auto Expo-Components 2023 held at Pragati Maidan, New Delhi
- Participation in Auto Expo 2022 held at Greater Noida, Uttar-Pradesh

Strengths:

- Honest, Team Player, Innovative
- Quick-Learner and Enthusiastic Person
- Pro-active, Hardworking

Hobbies:

- Dancing
- Reading Application Notes/White Papers.
- Learning New Technologies/Skills.

Personal Details:

- **Name:** Sachin Kumar
- **Father's Name:** Late Mr. Naresh Kumar
- **Languages:** English, Hindi
- **Passport:** Yes